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Simple, fast recovery, low-noise receiver amplifier for pulsed NMR experiments

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A fast recovery, low-noise receiver designed for use in pulsed NMR, especially high-resolution multiple-pulse NMR, is described. This receiver is easy to build from commercially available parts. Passive (but biased) voltage limiters placed before each amplifier in the cascade prevent any of the amplifiers from saturating, so it is only the limiters that ever saturate. Recovery times of 850 ns and 2.35 μ s are measured with a 50- Ω load and a high Q ($Q = 141$) resonant sample probe in the circuit, respectively. The receiver has a measured overall noise figure of 6.5 dB, including crossed diodes and filter on the input. An example of the use of the receiver to measure a short proton free-induction decay at 186.5 MHz is also shown.

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INTRODUCTION

One of the most challenging and important technical problems in pulsed NMR experimentation is building a receiver amplifier which can rapidly recover from saturation after severe overloading. In this paper, we present a low-noise, very fast recovery time receiver which is easy to build and can be used over a broad frequency range, including 186 MHz.

In pulsed NMR experiments, one applies to the sample one or more radio frequency (RF) pulses ranging from microseconds to milliseconds in length and $\sim 10^2$ W in amplitude. As soon as possible (ideally $< 1 \mu$ s) after the pulse, it is necessary to amplify a transient signal of $\sim 10^{-14}$ W coming from the sample. It is also highly desirable to have a low-noise figure for the receiver to increase the sensitivity of the spectrometer. There are two cases in particular in which it is crucial to have rapid recovery. One is the measurement of a short free-induction decay (FID) from a sample with a broad spectrum. In order to measure faithfully the spectrum, it is necessary to have a receiver which recovers from the RF pulse before the signal decays appreciably. The second case is in multiple-pulse NMR, where one applies sequences of short RF pulses and observes the magnetization of the sample in windows between these pulses. In order to achieve higher spectral resolution, it is necessary to use fast cycle times and to have a receiver which can recover in time to measure the signal between pulses.

There are actually three factors controlling the total dead time after an RF pulse in an NMR experiment. They are the quality factor (Q) of the transmitter or power amplifier, the Q of the resonant probe circuit, and the recovery time of the receiver amplifier. The obvious way to minimize transmitter ringdown is to use a broadband, low Q transmitter; however, other novel approaches have been taken.¹ The Q of the probe represents a more difficult problem because high Q is desirable

for a good signal-to-noise ratio and a large RF magnetic field, while low Q is desirable for a fast ringdown and shorter dead time. A variety of ingenious methods has been used to deal with the probe problem.²⁻⁴ The recovery time of the receiver is probably the most important and challenging of the three. By merely choosing standard low-noise amplifiers without considering recovery time, one is likely to have a dead time of hundreds of microseconds following a high-power RF pulse. Receivers in commercial NMR spectrometers also have dead times which are long enough to preclude very fast multiple pulse sequences. Many workers have approached the receiver recovery time problem with varying degrees of success.⁴⁻¹¹ Techniques involving active gating of the receiver input have been tried, but the overall most successful techniques seem to involve passive voltage limiters incorporated at some point in the circuit.⁸⁻¹⁰ Adduci *et al.*¹¹ have the best receiver recovery time in the literature to date by using limiting amplifiers, which presumably have some built-in voltage limiters.

In this paper, we describe a receiver built from commercially available components. This receiver has a very rapid recovery from saturation, a very good noise figure (6.5 dB), large bandwidth (50–500 MHz), and it works well at high frequencies. (We use it routinely at 186.5 and 199 MHz).

I. DETAILS AND PERFORMANCE

The receiver is shown in a block diagram in Fig. 1. First, the input is protected from destruction by crossed diodes to ground (actually two pairs of 1N914 fast-switching diodes). The input comes from the probe and transmitter via a 50- Ω , quarter-wavelength cable in the configuration¹² utilized in most pulsed NMR spectrometers. The diodes are followed in Fig. 1 by a bandpass filter (BP) to eliminate unwanted low- and high-frequency transients which would cause unnecessary overdriving of

the receiver. (Any good bandpass filter should do the job well. We happen to use a 4-pole bandpass filter with a center frequency of 180 MHz, a 20 MHz bandwidth, and 50 Ω impedance, Cir-Q-Tel¹³ Model FBT/2-180/20-4/50-3A/3A). The bandwidth is centered near the resonance frequency (in our case 186.5 MHz), so there is only about 0.9 dB insertion loss at the signal frequency, thus causing an increase of 0.9 dB in the noise figure. This bandpass filter is the only nonbroadband component in the receiving system. Since it is external to the actual receiver limiters and amplifiers, and it is in a separate coaxial package, it can be changed in a matter of seconds. There is no tuning whatsoever, so the receiver is essentially broadbanded and can be set for any resonant frequency by merely substituting the appropriate bandpass filter.

After the filter comes the main part of the receiver (see Fig. 1), which consists of a series of six cascaded sections, each composed of a voltage limiter (L) and an amplifier (A). The limiters are Watkins-Johnson¹⁴ Model WJ-L1, and they have a bandwidth of 50–1,000 MHz. Each limiter consists of crossed diodes to ground with a series 50 pf capacitor on the input and the output with a diode bridge array in between. Schottky barrier diodes are used and the design is symmetric so that symmetric clipping occurs. The absolute limiting voltage may be chosen from –6 to 0 dBm by adjusting the bias voltage. We operate the limiters at +10 V bias which limits the output to –2 dBm maximum. At this level the insertion loss is typically 2.2 dB. The recovery time of each limiter from saturation is about 5 ns typically. The gain of each amplifier is chosen so that the output from the previous limiter will not drive the amplifier into saturation. In fact, all the stages are chosen so that the output of each amplifier is well below its 1 dB compression point.

Since the first stage of amplification primarily determines the noise figure of the whole cascade, it was necessary to choose the lowest noise figure first stage that would not be overdriven by –2 dBm input. We chose for the first two amplifier stages (A1 and A2 in Fig. 1) the Avantek¹⁵ Model UTO-544, which has a bandwidth of 10–500 MHz, noise figure 2.5 dB typically, a gain of 11.5 dB, and a 1 dB compression point of +13 dBm at 200 MHz. The last four stages of the cascade (A3–A6) were chosen to provide greater gain, but also without saturation or compression. These four stages are

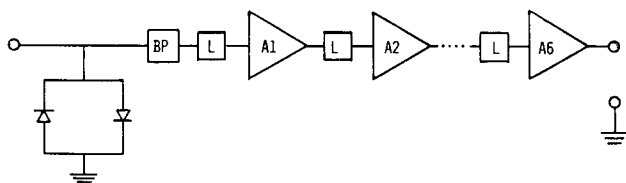


FIG. 1. Block diagram of receiver amplifier. The input to the receiver is protected by crossed diodes to ground. Next, a bandpass filter (BP) is used to filter out low-frequency transients. Following this are six sets of voltage limiters (L) and amplifiers (A1–A6). The input to the receiver comes through a typical quarter-wave length configuration.

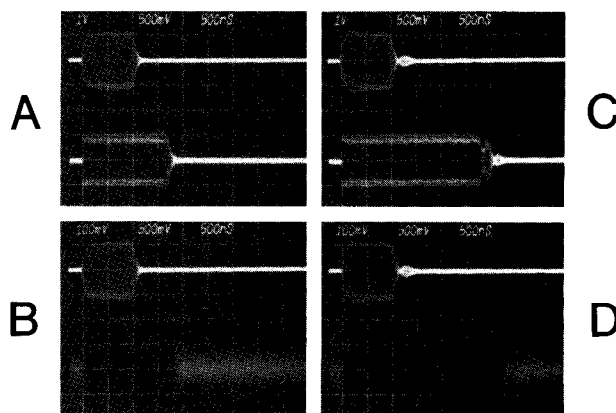


FIG. 2. Oscilloscope photographs illustrating the rapid recovery of the receiver. In all four figures, (A)–(D), the top trace is the 186.5 MHz pulse which is saturating the receiver and being viewed with a directional coupler, while the bottom trace is the direct output from the receiver. Figures (A) and (B) were taken with a 50 Ω resistive load placed in lieu of a high Q tuned resonant probe. Figure (B) has the receiver output magnified by a factor of ten to show more clearly the end of saturation and return of the ambient noise floor. Figures (C) and (D) were taken with a high Q ($Q = 141$) tuned resonant probe in the circuit. Again Fig. (D) has been magnified by a factor of ten. In all four figures, the horizontal scale is 500 ns/div and the vertical scale for the top trace is 500 mV/div. (It is necessary to multiply the oscilloscope voltage for the RF pulse by a factor of 125 to compensate for attenuation in our dual directional coupler). In Figs. (A) and (C), the bottom trace has a vertical scale of 1 V/div while Figs. (B) and (D) have 100 mV/div.

all Watkins-Johnson¹⁴ Model WJ-A57. This module has a bandwidth of 5–600 MHz, noise figure of 4.6 dB, gain of 14.5 dB, and a 1 dB compression point of 14.2 dBm at 200 MHz. The limiters and amplifiers are all of the TO-8 can configuration which can be mounted on micro-strip circuit boards (Avantek¹⁵ TB Series) within special metal cases (Avantek¹⁵ TC Series). Excluding the coaxial bandpass filter on the input, the bandwidth of the final cascade is 50–500 MHz. The lower end is determined by the voltage limiters, but we have recently brought other voltage limiters from Avantek,¹⁵ Model UTL-1002 which are claimed to have a bandwidth of 5–1,000 MHz. We are incorporating these in a receiver to be used at 20 MHz and do not anticipate any difficulties since the major difference in these limiters is the size of the input and output capacitors. It should also be noted here that Avantek¹⁵ makes a limiter Model UTL-1001 which has specifications virtually identical to the Watkins-Johnson¹⁴ Model WJ-L1 and could probably be substituted for it.

Figure 2 shows oscilloscope photographs which illustrate the performance of the receiver. The manner of testing the recovery time of the receiver was to apply a 1- μ s (actually a 60 W, 90° NMR pulse), 186.5 MHz pulse to the actual NMR high Q probe ($Q = 141$) and also to a 50- Ω load in place of the probe. The output of the receiver was then monitored. Figures 2(A) and (B) show the case of the 50 Ω load, with the top trace in each being the actual RF pulse viewed through a dual-directional coupler and the bottom trace being the direct output of the receiver. Figure 2(B) is the same as (A), except that the receiver output has been magnified by a factor of ten to make it easier to determine the recovery

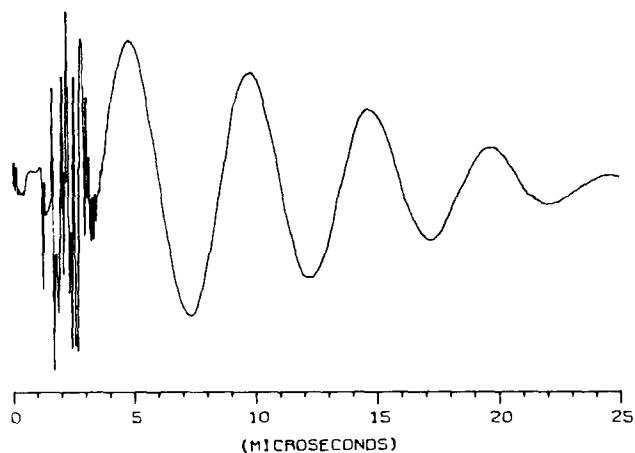


FIG. 3. Free-induction decay (FID) of ^1H in $\text{Ca}(\text{OH})_2$ showing the rapid recovery of the receiver in actual use. A $1\text{ }\mu\text{s}$, 90° pulse is applied to the sample at time, $t = 0$. Less than $2.4\text{ }\mu\text{s}$ after the end of the pulse the receiver has recovered and the $\text{Ca}(\text{OH})_2$ signal is being amplified without distortion. About $1.6\text{ }\mu\text{s}$ of the recovery time is due to the ringing of the sample probe tuned circuit ($Q = 141$).

time. From Figs. 2(A) and (B), we can see that the receiver noise floor has recovered within 850 ns after the end of the RF pulse. In Figs. 2(C) and (D), we see the case where the high Q probe is used. Again the top trace is the RF pulse while the bottom is the receiver output, and Fig. 2(D) is magnified by a factor of ten over Fig. 2(C). From these photographs, we see that the receiver noise floor has returned in less than $2.35\text{ }\mu\text{s}$. Most of this $2.35\text{ }\mu\text{s}$ is due to the longer ringing of the high Q probe, but this is still fast enough to use very fast cycle times in high-resolution multiple pulse experiments. A lower Q probe should give a recovery time which is somewhere between $2.35\text{--}0.85\text{ }\mu\text{s}$.

We have found it advantageous to place a home-built variable bandpass filter on the input of our broadband RF transmitter, so that we can tune the pulse transients in multiple pulse experiments. Because of this fact, our transmitter has a narrower effective bandwidth and thus the RF pulse takes longer to turn on and off. This is clear in Figs. 2(A) and (B) where the RF pulse (top trace) has a transient time constant of about 100 ns . We therefore strongly feel that some large part of the 850 ns recovery time is, in fact, due to the effectively higher Q of the transmitter. We suggest that perhaps one half of the time may be attributed to the transmitter, thus making the real recovery time of the amplifier less than $0.5\text{ }\mu\text{s}$.

In order to verify that the recovery time as measured by looking at the noise floor recovery of the receiver is in fact the same as that in a real experimental situation, we have taken the data shown in Fig. 3. This figure shows a short proton FID taken from a powdered sample of $\text{Ca}(\text{OH})_2$ at 186.5 MHz . The strong homogeneous dipolar interaction makes the spectrum about 45 KHz wide and the FID quite short. A 90° pulse of $1\text{ }\mu\text{s}$ was used, and the receiver has apparently totally recovered and begun to amplify the proton magnetization in less than $3.4\text{ }\mu\text{s}$ from time, $t = 0$, which was chosen to be

the beginning of the RF pulse. Thus, our recovery time of $2.4\text{ }\mu\text{s}$ from the FID seems to agree well with the $2.35\text{ }\mu\text{s}$ reported earlier in this paper.

The noise figure of the receiver cascade was measured using a Hewlett-Packard¹⁶ Model 141 T Spectrum Analyzer according to the procedures described in the Hewlett-Packard Applications Note 150-4.¹⁷ We measured a noise figure of 5.1 dB for the total cascade of six limiters and amplifiers, which agrees within 0.1 dB of what one would calculate theoretically given the manufacturer's specifications and typical performance curves for the various components. The 2.5 dB noise figure of the initial amplifier stage is degraded by 2.2 dB due to the insertion loss of the first limiter. The remaining 0.4 dB of the noise figure are contributed by the subsequent stages of the cascade. The measured insertion loss of 0.9 dB for the bandpass filter and 0.5 dB for the crossed diodes give a total of 6.5 dB noise figure for the receiver as shown in Fig. 1. We also measured the gain of the cascade to be 68 dB , thus the gain of the receiver with crossed diodes and bandpass filter is 66.6 dB . From Fig. 2 it is evident that with the limiters the dynamic range is only about 26 dB . However, this can be effectively increased to over 70 dB , if necessary, without appreciably degrading the noise figure, by placing attenuators between the later stages of the receiver or bypassing them altogether. Furthermore, audio filtering after phase sensitive detection effectively enhances dynamic range by a large factor.

II. DISCUSSION

The receiver described here has many advantages over other NMR receivers, either commercial or home built. The measured recovery time is better than or comparable to any we could find in the literature. The recovery time of 850 ns that we measured with a $50\text{-}\Omega$ load is, perhaps, even artificially long due to the ringing of the transmitter. Unlike the receiver described by Adduci *et al.*,¹¹ our receiver's recovery time does not increase with the length of the RF pulse, and has been measured to be the same with pulses up to 100 ms . The receiver was also designed with the goal of minimizing the noise figure, which was measured to be 6.5 dB for the whole receiver, including the crossed diodes and the bandpass filter on the input. The only narrow-banded component in the system is the coaxial bandpass filter which can be changed in seconds, thus making the same receiver adequate for many different nuclei. By choosing the correct limiters and amplifiers, one could potentially cover the entire bandwidth from $5\text{--}1,000\text{ MHz}$. The gain of 66.6 dB for the whole receiver has been adequate for all of our needs to date. The simplicity of assembling readily available and relatively inexpensive commercial components makes this receiver quite attractive. By having a modular system, it is quite easy to tailor a receiver for a specific need or experiment within minutes. Since the protection and recovery of the receiver do not in-

volve active gating, there are none of the complications due to complex timing of the gates and to the inevitable gating transients. This receiver has greatly enhanced our capability to do experiments involving short FID's and multiple-pulse NMR, as well as being far more than adequate for less demanding experiments, thus making it a useful all-purpose receiver.

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- * This work was supported by the U.S. Department of Energy.
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